# Gayatri Vidya Parishad College of Engineering Autonomous

Data Science | Course Code: DS301

## Applied Statistics & Data Science - Comprehensive Syllabus Lecture Notes

Syllabus Focus: DS301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Applied Statistics & Data Science. Includes d

### I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

**Unit 1: Probability Distributions & Sampling Theory**

" Syllabus Scope: Discrete and continuous random variables, Normal, Binomial, Poisson, Exponential distri

**Unit 2: Parametric Hypothesis Testing**

" Syllabus Scope: Null & alternative hypotheses, Type I and Type II errors, p-value interpretation, Z-tes

# DS301: Applied Statistics & Data Science - Advanced Topics & Assessment

Academic Track: Data Science - Units 3, 4, 5 & Examination Matrix

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## II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

**Unit 3: Categorical Analysis & Chi-Square Tests**

" Goodness-of-fit tests, Test of independence in contingency tables, Contingency coefficients, Degrees of free

**Unit 4: Analysis of Variance ANOVA**

" One-way and Two-way ANOVA, F-distribution, Total sum of squares decomposition, Post-hoc tests Tukey HSD

**Unit 5: Correlation & Multivariate Regression**

" Pearson and Spearman rank correlation, Multiple linear regression, Multicollinearity, Variance Inflation Fac

## III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Formulate a two-tailed Student t-test for comparing independent sample means with unequal variances We
2. Construct an ANOVA table for a single-factor experiment with 4 treatment groups and calculate the F-st
3. State and prove the Central Limit Theorem and discuss its application in large-sample statistical infe

## IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Douglas C. Montgomery - Applied Statistics and Probability for Engineers Wiley 7th Ed
* Gareth James et al. - An Introduction to Statistical Learning Springer

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